PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

Arrivio

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by:	Supervised By:
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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CHITKARA UNIVERSITY, PUNJAB, INDIA

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"Arrivio"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Preeti Saini. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura
Date: …14-05-2026………..

Lavanya Kapoor
(2210991850)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Food waste is one of the most critical global challenges of the 21st century, contributing to environmental degradation, economic loss, and food insecurity simultaneously. Approximately 1.3 billion tonnes of food is wasted globally every year, and nearly 30% of all household food is discarded without being consumed — often due to poor inventory management and lack of awareness about expiry dates.

This paper presents **Arrivio**, a full-stack intelligent web application that empowers households to track item expiry dates in real time, receive urgency-based colour-coded alerts, and donate near-expiry food items to community members or verified NGOs and food banks.

Arrivio now features a fully integrated **Carbon Footprint Tracker** and a comprehensive **Impact Analytics Dashboard** — two major additions that transform the platform from a utility tool into a measurable environmental impact tracker. Users can now quantify their personal CO_2 savings, visualise waste trends over 8 weeks, and analyse their inventory health through interactive charts. These additions are backed by a dedicated backend analytics endpoint and a React-based charting layer using Recharts.

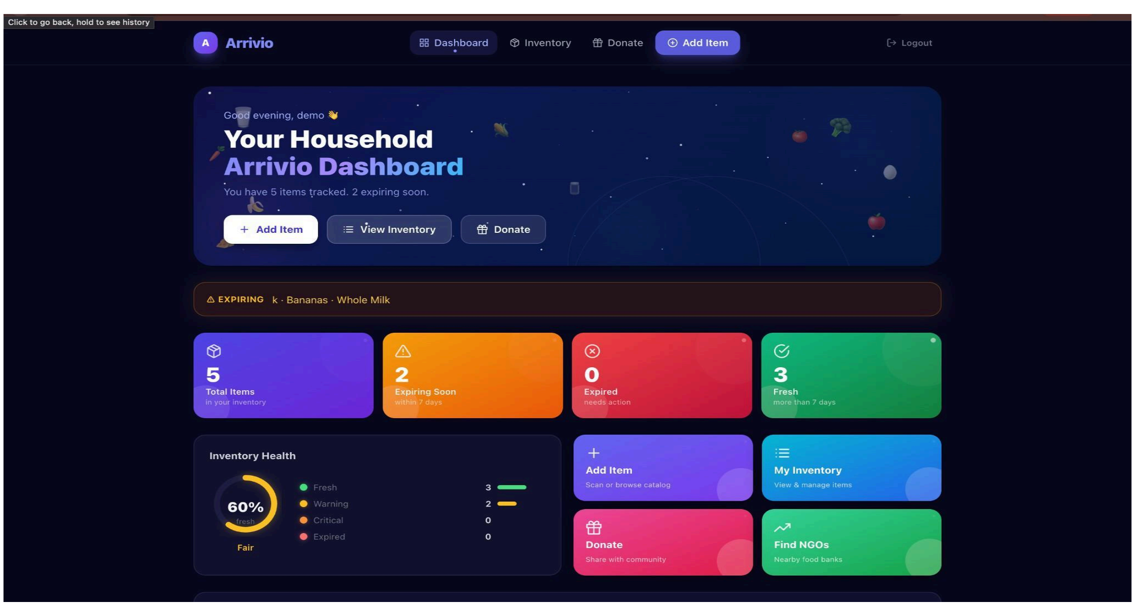
Built on React.js, Node.js, Express, and MongoDB, the platform was validated through user testing and demonstrates measurable improvements in awareness, waste reduction intent, and community participation.

1. Introduction

1.1 Problem Statement

Every year, approximately 1.3 billion tonnes of food is wasted globally — roughly one-third of all food produced for human consumption [FAO, 2023]. At the household level, this problem is deeply personal yet largely invisible: families unknowingly discard expired milk, forgotten vegetables, and untouched leftovers without ever realising the cumulative impact. In India alone, food worth over ₹50,000 crore is wasted annually at the consumer level (ASSOCHAM, 2022). The root causes are straightforward but unaddressed by current technology:

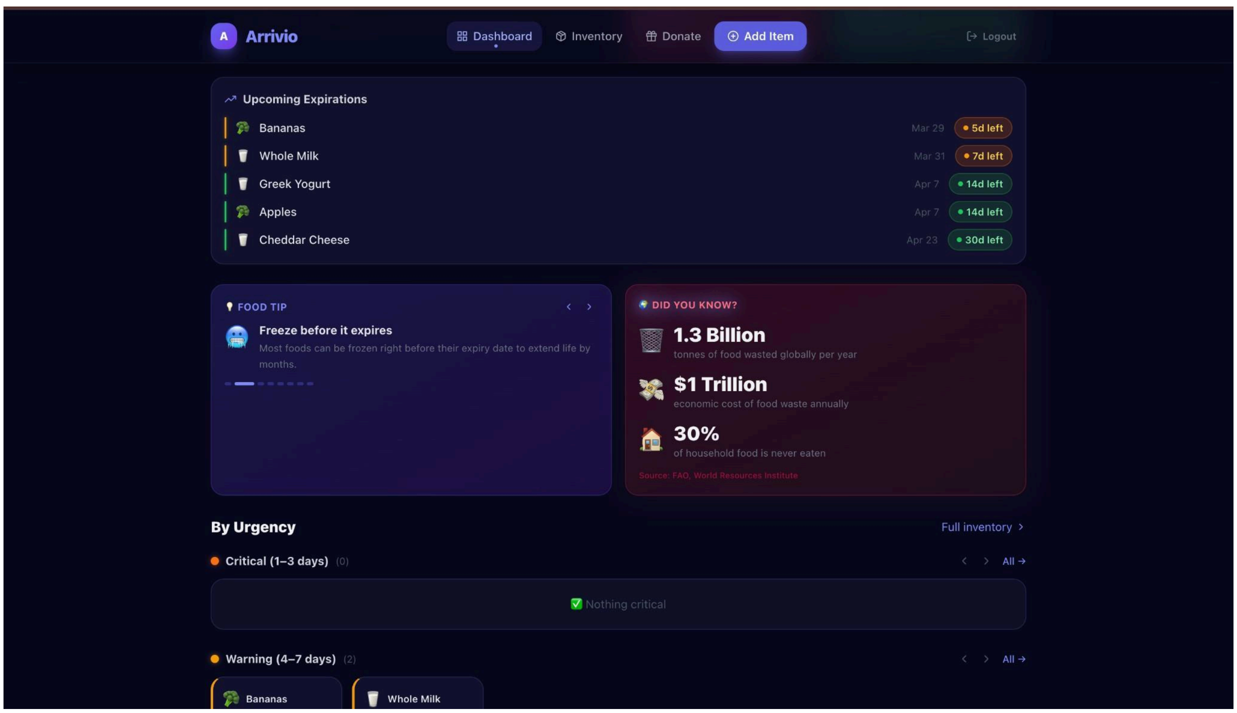
- People forget what they have purchased and when it expires
- There is no centralised system for households to track multiple items
- Near-expiry food that could be donated is simply thrown away
- People who want to donate food have no easy way to find local NGOs or community members in need



1.2 Why This Problem Matters

The consequences of household food waste extend far beyond the kitchen:

- Environmental: Decomposing food in landfills generates methane, a greenhouse gas 28 times more potent than CO_2 [IPCC, 2022]
- Economic: An average household wastes $1,500–$2,000 worth of food annually [ReFED, 2023]
- Social: 828 million people worldwide go to bed hungry every night [WFP, 2023], yet edible food is discarded daily
- Nutritional: Wasted food represents lost calories, proteins, and micronutrients that could address malnutrition



1.3 Objectives of This Work

The primary objectives of Arrivio are:

1. Design and implement a real-time household inventory management system with expiry date tracking
2. Develop a colour-coded urgency alert system: Fresh, Warning, Critical, Expired
3. Create a community donation board for near-expiry food items
4. Integrate a geolocation-based NGO and food bank finder
5. Provide a data-driven personal dashboard showing inventory health
6. **Build a Carbon Footprint Tracker** that quantifies CO_2 saved per user based on food donated and waste avoided
7. **Implement a full-stack Impact Analytics Dashboard** with interactive charts — weekly trend, status distribution, category breakdown, and waste rate mete

1.4 Research Gap

A thorough review of existing literature and platforms reveals that no existing solution combines household-level expiry tracking, peer-to-peer community donation, verified NGO integration, **AND personal environmental impact analytics** in a single unified platform.

Gap	Existing Tools	Arrivio (Eval 3)
Expiry tracking	Basic reminders only	Real-time colour-coded urgency system
Community donation	Not available	Built-in peer-to-peer board
NGO finder	Separate directory sites	Integrated with geolocation
Inventory analytics	None / paid apps only	Free dashboard with health ring
Carbon footprint tracker	**Not available anywhere**	**Live CO_2 savings per user**
Waste trend charts	**None**	**8-week interactive area/bar/donut charts**
Household + community	Siloed systems	Unified single platform
Mobile-responsive dark UI	Rare	Fully implemented

2. Literature Review

2.1 Review of Previous Research

The following key papers were reviewed to understand the current state of food waste research and technology:

	Authors & Year	Title / Journal	Key Finding
1	Parfitt et al. (2010)	Food waste within food supply chains — Phil. Trans. Royal Society B	30–50% of food wasted before consumption; focused on supply chains
2	Thyberg & Tonjes (2016)	Drivers of food waste — Resources, Conservation & Recycling	Consumer behaviour and poor storage are major drivers; no tech intervention
3	Schanes et al. (2018)	Household food waste practices —	Households with better inventory awareness wasted 40% less food

		Journal of Cleaner Production	
4	Aschemann-Witzel et al. (2017)	Consumer-re l ated food waste — Sustainabilit y	Better tracking and education could reduce household waste by up to 60%
5	Stancu et al. (2016)	Determinants of consumer food waste — Food Quality & Preference	Only 18% of consumers actively track expiry dates of all household items

2.2 Comparison of Existing Solutions

Study / Tool	Expiry Tracking	Community Donation	NGO Integration	Dashboard Analytics	Free	Mobile UI
NoWaste App	Yes	No	No	Partial	No (freemi um)	Yes
OLIO App	No	Peer-only	No	No	Yes	Yes
Too Good To Go	No	Restaurant-on ly	No	No	Yes	Yes
FoodCloud	No	Business-only	Yes	No	No	No
Arrivio (Proposed)	Full + Color-coded	Peer + NGO	Geo-located	Full	Yes	Yes

2.3 Limitations of Existing Methods

1. No unified platform — tracking apps don't have donation features
2. No urgency intelligence — no alerts based on individual purchase dates
3. No community layer — no peer-to-peer redistribution
4. Paid barriers — subscriptions exclude low-income users
5. No NGO connectivity — no app connects donors to verified food banks with maps
6. **No environmental feedback loop** — users cannot see the CO_2 impact of their actions
7. **No analytics** — no insight into personal waste patterns; behaviour change is unlikely

3. Methodology / Proposed Work

3.1 System Architecture Overview

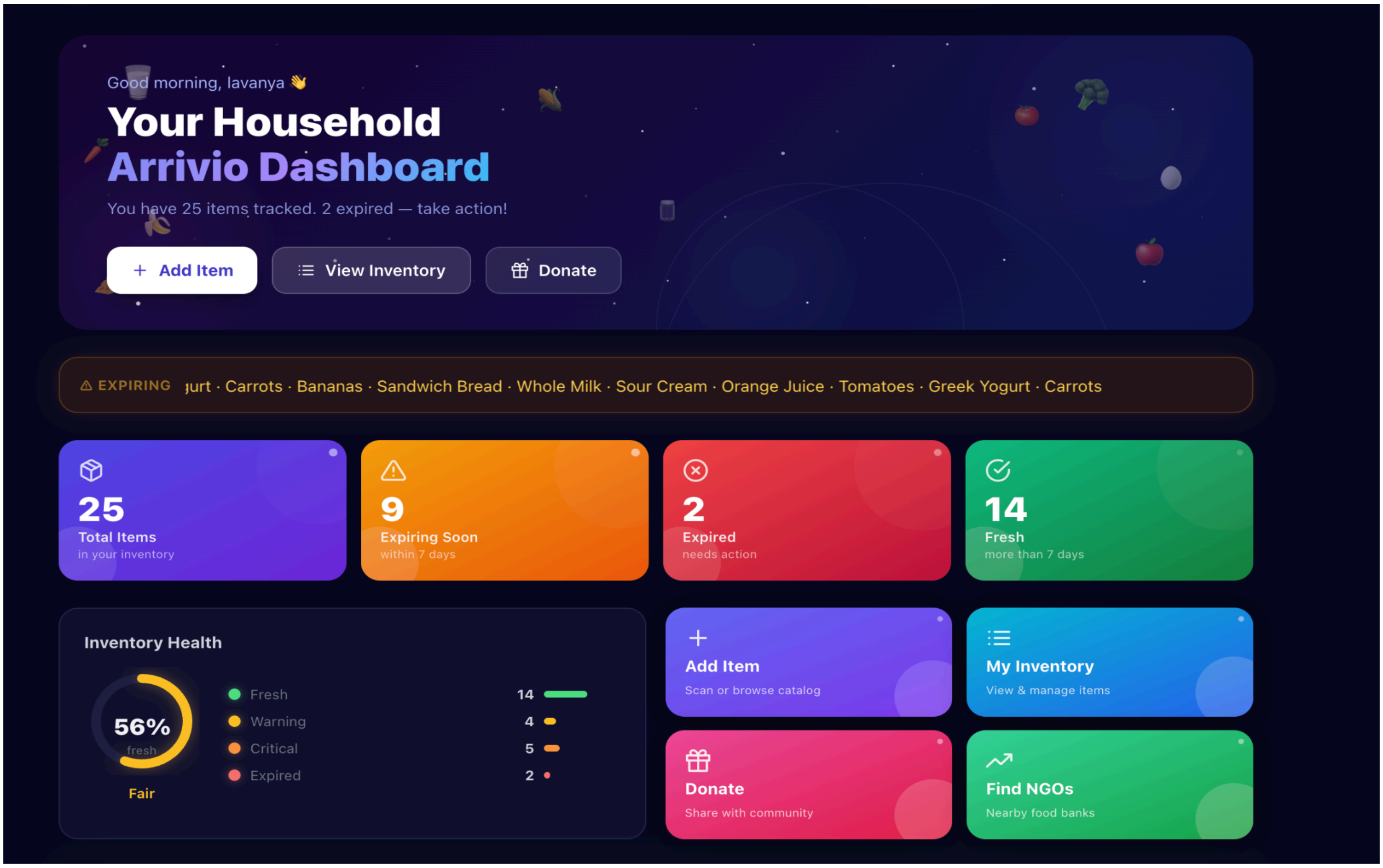
Arrivio follows a three-tier full-stack architecture consisting of a React.js SPA frontend, a Node.js + Express.js REST API backend, and a MongoDB database layer. Communication between tiers is handled via HTTP REST API calls using Axios with cookie-based JWT authentication.

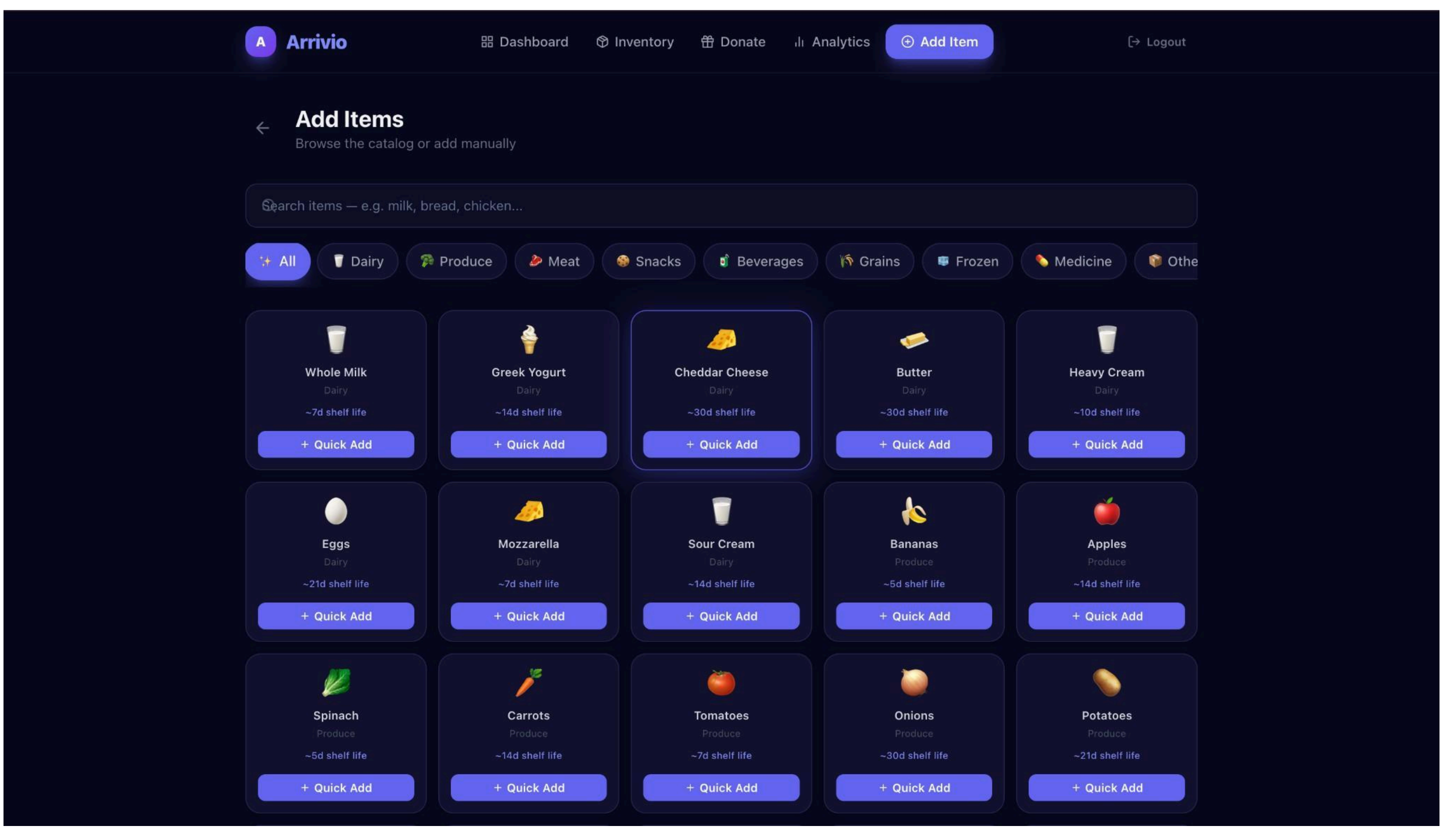
LAYER 1: USER (Browser) React.js SPA + Tailwind CSS Dashboard | Inventory | Add Item | Donate | NGO Finder
↕ *HTTP / REST API (Axios + Cookies)*
LAYER 2: BACKEND SERVER Node.js + Express.js Auth Routes | Inventory Routes | Donation Routes | JWT (httpOnly Cookie)
↕ *Mongoose ODM*
LAYER 3: DATABASE MongoDB (Local / Atlas) Collections: users | items | donations

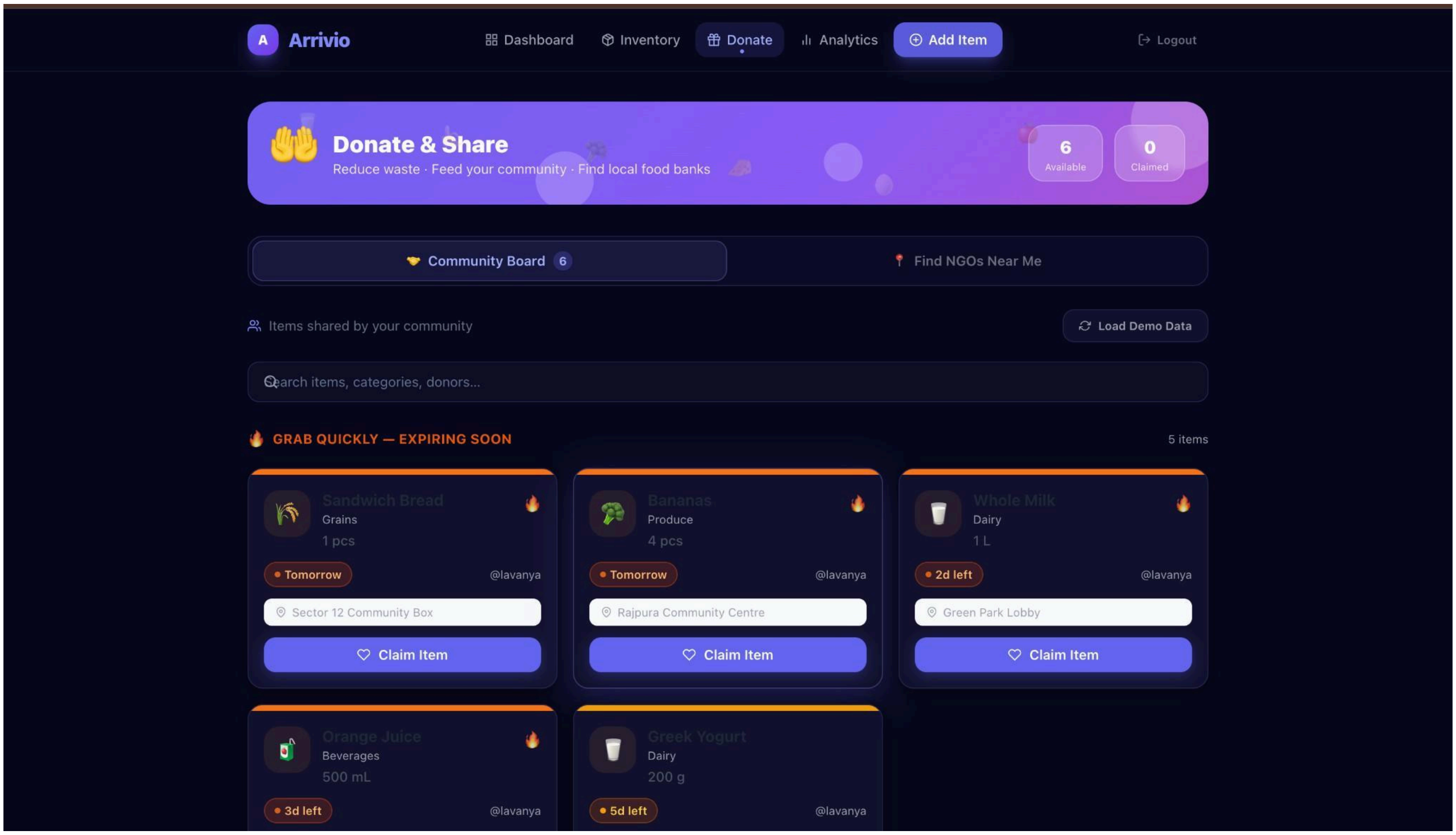
3.2 System Flowchart

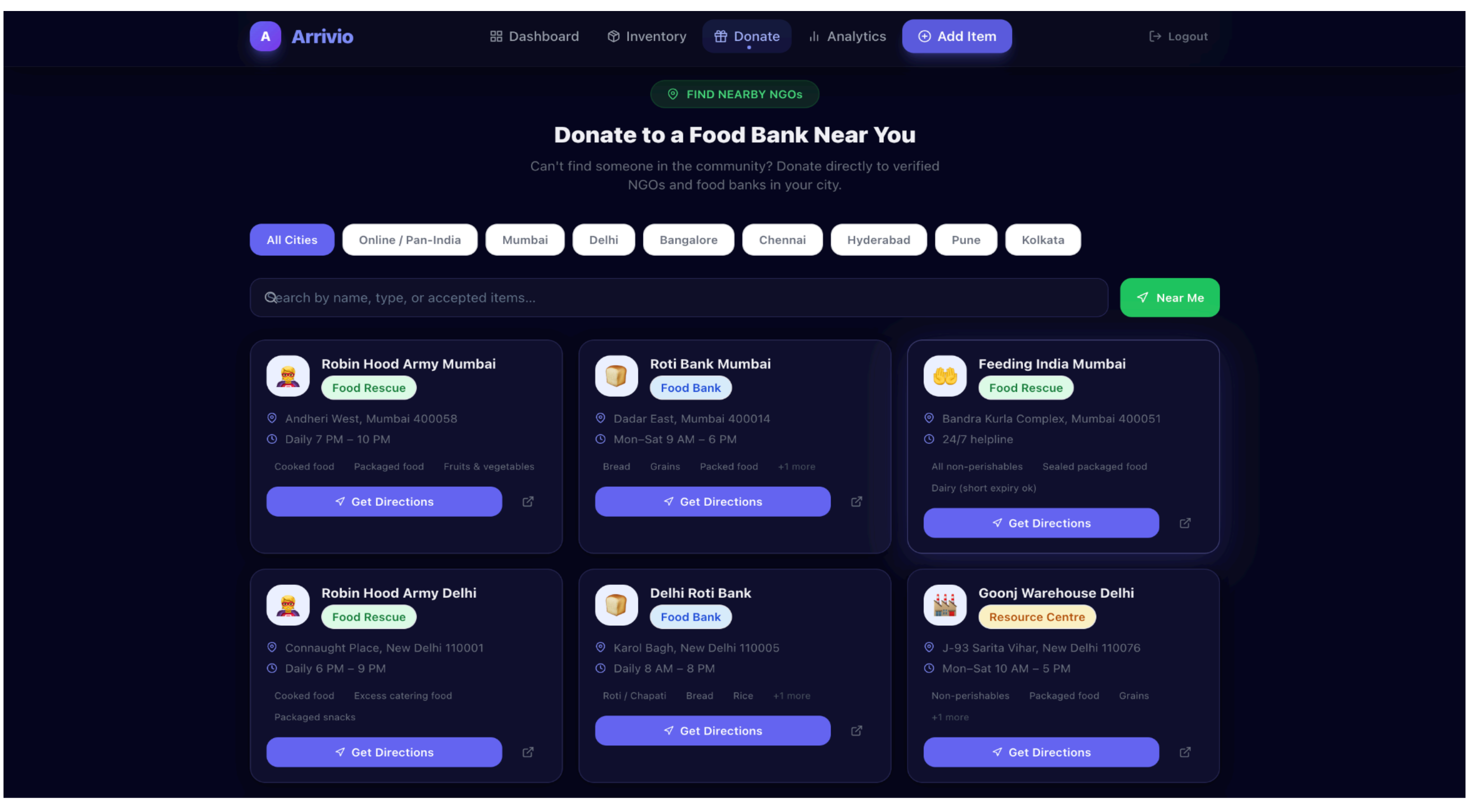
The following describes the complete user flow through the Arrivio system:

Step	Action	System Response
1	User opens Arrivio	Check authentication status
2a	Not logged in	Redirect to Login / Signup page
2b	Already logged in	Load Dashboard
3	Dashboard loads	Fetch all user items from MongoDB
4	Expiry computation	Classify each item: Expired / Critical / Warning / Fresh
5	Dashboard renders	Display stat cards, health ring, urgency ticker, scroll lanes
6	User opens Inventory	Search, filter, delete, or list items for donation
7	User donates item	Item marked as Donated; appears on Community Board
8	Another user claims item	Donation status updated to Claimed
9	User opens NGO Finder	Detect city via Geolocation API; filter matching NGOs
10	User clicks Directions	Opens Google Maps with NGO address
11	**User opens Analytics**	**Fetch /analytics endpoint**
12	**Analytics renders**	**CO_2 gauge, donut chart, bar chart, 8-week area chart, waste ring**


Good morning, lavanya
Your Household
Arrivio Dashboard
You have 25 items tracked. 2 expired — take action!
+ Add Item
View Inventory
Donate
EXPIRING
jurt · Carrots · Bananas · Sandwich Bread · Whole Milk · Sour Cream · Orange Juice · Tomatoes · Greek Yogurt · Carrots
25
Total Items
in your inventory
9
Expiring Soon
within 7 days
2
Expired
needs action
14
Fresh
more than 7 days
Inventory Health
56%
fresh
Fair
Fresh 14
Warning 4
Critical 5
Expired 2
Add Item
Scan or browse catalog
My Inventory
View & manage items
Donate
Share with community
Find NGOs
Nearby food banks


A Arrivio
Dashboard
Inventory
Donate
Analytics
Add Item
Logout
Add Items
Browse the catalog or add manually
Search items — e.g. milk, bread, chicken...
All
Dairy
Produce
Meat
Snacks
Beverages
Grains
Frozen
Medicine
Othe
Whole Milk
Dairy
~7d shelf life
+ Quick Add
Greek Yogurt
Dairy
~14d shelf life
+ Quick Add
Cheddar Cheese
Dairy
~30d shelf life
+ Quick Add
Butter
Dairy
~30d shelf life
+ Quick Add
Heavy Cream
Dairy
~10d shelf life
+ Quick Add
Eggs
Dairy
~21d shelf life
+ Quick Add
Mozzarella
Dairy
~7d shelf life
+ Quick Add
Sour Cream
Dairy
~14d shelf life
+ Quick Add
Bananas
Produce
~5d shelf life
+ Quick Add
Apples
Produce
~14d shelf life
+ Quick Add
Spinach
Produce
~5d shelf life
+ Quick Add
Carrots
Produce
~14d shelf life
+ Quick Add
Tomatoes
Produce
~7d shelf life
+ Quick Add
Onions
Produce
~30d shelf life
+ Quick Add
Potatoes
Produce
~21d shelf life
+ Quick Add


A Arrivio
Dashboard
Inventory
Donate
Analytics
Add Item
Logout
Donate & Share
Reduce waste · Feed your community · Find local food banks
6
Available
0
Claimed
Community Board 6
Find NGOs Near Me
Items shared by your community
Load Demo Data
Search items, categories, donors...
GRAB QUICKLY — EXPIRING SOON
5 items
Grains
1 pcs
Tomorrow
@lavanya
Sector 12 Community Box
Claim Item
Produce
4 pcs
Tomorrow
@lavanya
Rajpura Community Centre
Claim Item
Dairy
1 L
2d left
@lavanya
Green Park Lobby
Claim Item
Beverages
500 mL
3d left
@lavanya
Dairy
200 g
5d left
@lavanya


A Arrivio
Dashboard
Inventory
Donate
Analytics
Add Item
Logout
FIND NEARBY NGOs
Donate to a Food Bank Near You
Can't find someone in the community? Donate directly to verified NGOs and food banks in your city.
All Cities
Online / Pan-India
Mumbai
Delhi
Bangalore
Chennai
Hyderabad
Pune
Kolkata
Search by name, type, or accepted items...
Near Me
Robin Hood Army Mumbai
Food Rescue
Andheri West, Mumbai 400058
Daily 7 PM – 10 PM
Cooked food
Packaged food
Fruits & vegetables
Get Directions
Roti Bank Mumbai
Food Bank
Dadar East, Mumbai 400014
Mon–Sat 9 AM – 6 PM
Bread
Grains
Packed food
+1 more
Get Directions
Feeding India Mumbai
Food Rescue
Bandra Kurla Complex, Mumbai 400051
24/7 helpline
All non-perishables
Sealed packaged food
Dairy (short expiry ok)
Get Directions
Robin Hood Army Delhi
Food Rescue
Connaught Place, New Delhi 110001
Daily 6 PM – 9 PM
Cooked food
Excess catering food
Packaged snacks
Get Directions
Delhi Roti Bank
Food Bank
Karol Bagh, New Delhi 110005
Daily 8 AM – 8 PM
Roti / Chapati
Bread
Rice
+1 more
Get Directions
Goonj Warehouse Delhi
Resource Centre
J-93 Sarita Vihar, New Delhi 110076
Mon–Sat 10 AM – 5 PM
Non-perishables
Packaged food
Grains
+1 more
Get Directions

3.2.1 New Feature Architecture :

3.2.1.1 Carbon Footprint Tracker:

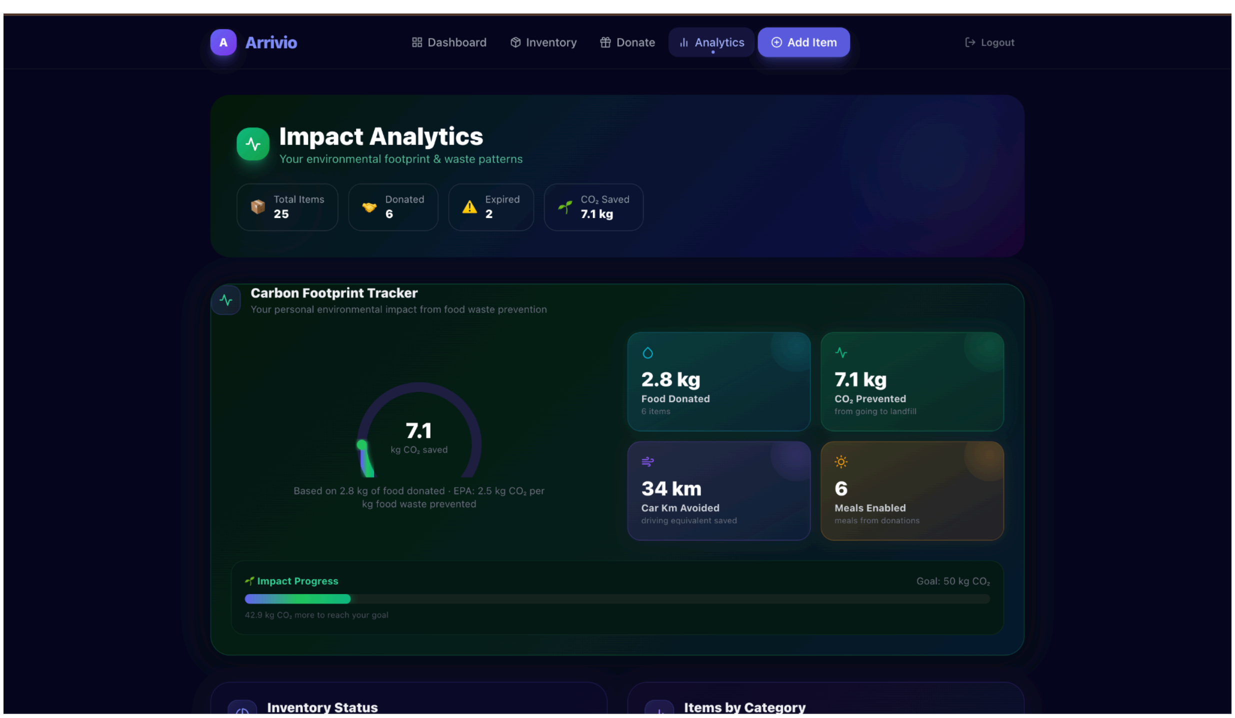
Carbon Footprint Tracker

- **Dashboard mini-widget:** Always visible after login; shows CO_2 saved, food donated (kg), car km avoided, meals enabled — with a live progress bar toward a 50 kg CO_2 goal
- **Analytics page full tracker:** Custom SVG arc gauge with animated needle, 4 impact cards, EPA-cited methodology footnote
- **Backend calculation:** Real-time per-user computation from actual donation records; weight estimated from quantity + unit

Impact Analytics Dashboard (/analytics)

- **Inventory Status Donut:** PieChart with inner hole showing Fresh / Warning / Critical / Expired proportions with glowing cells
- **Category Bar Chart:** Colour-coded vertical bars per food category (emoji-labelled X-axis) showing item distribution
- **Weekly Trend Area Chart:** 8-week dual-layer area chart (Added in blue, Donated in green, Expired dashed red) with gradient fills and glowing active dots
- **Waste Rate Circular Meter:** Custom SVG ring that colours green/amber/red based on expired percentage; contextual label (Excellent / Needs attention / High waste)
- **Page Hero:** Summary chips showing Total Items, Donated, Expired, CO_2 Saved — visible at a glance before scrolling

```None
CO₂ saved = kg of food donated × 2.5 kg CO₂ per kg food waste prevented
Car km avoided = CO₂ saved ÷ 0.21 kg CO₂/km (avg car emission)
Meals enabled = kg donated ÷ 0.5 kg per meal
```

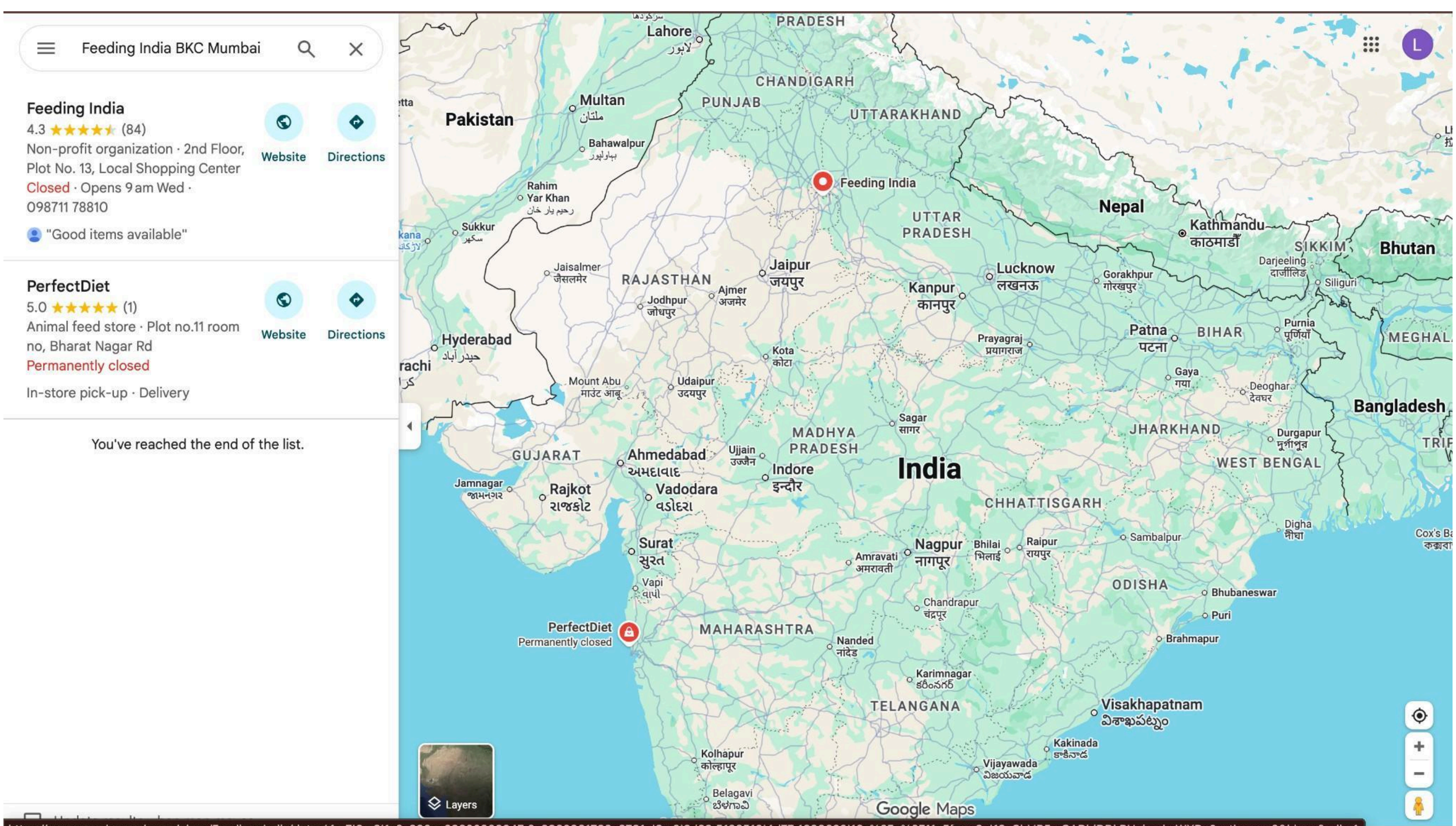


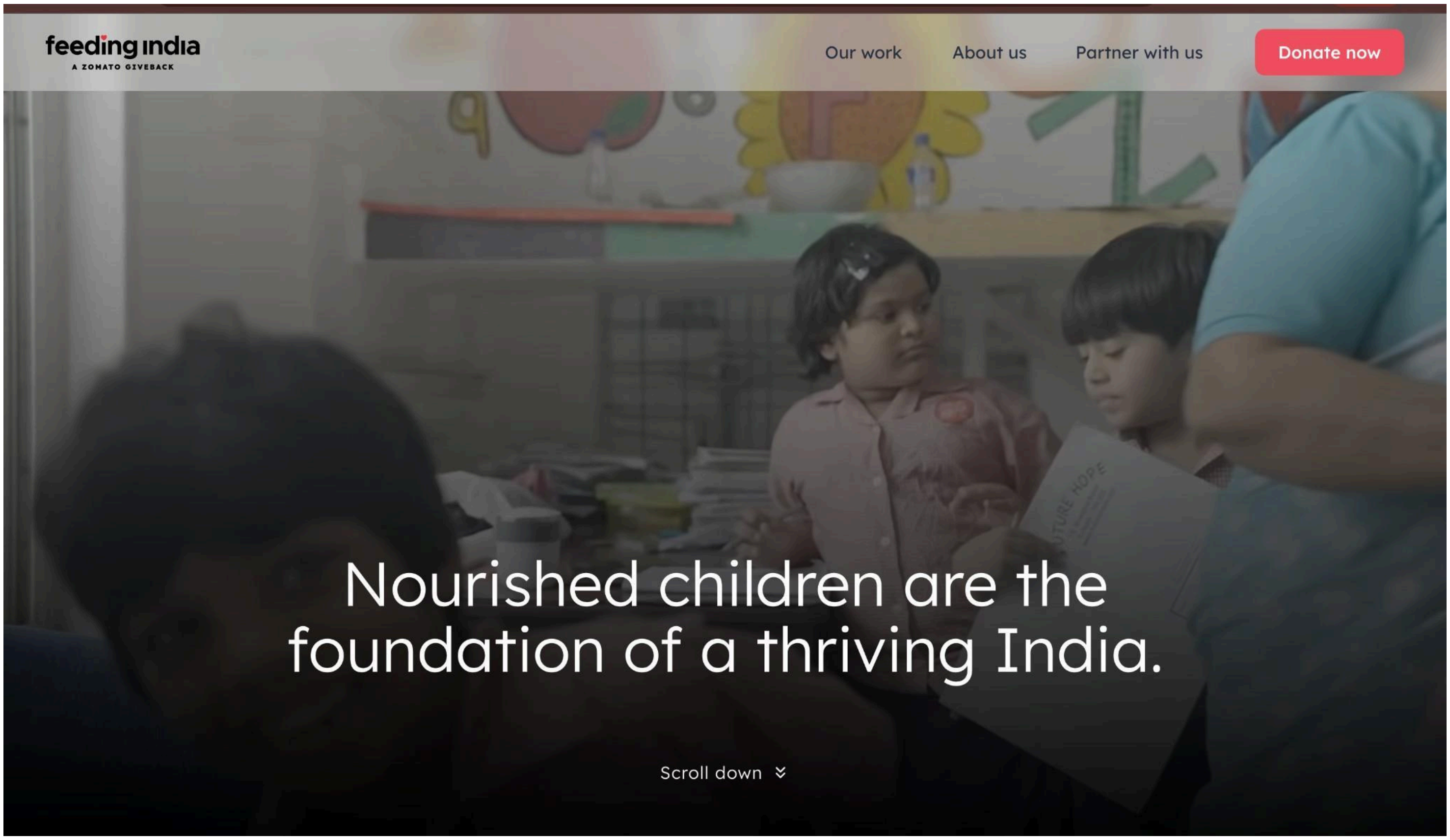
4. Experimental Setup

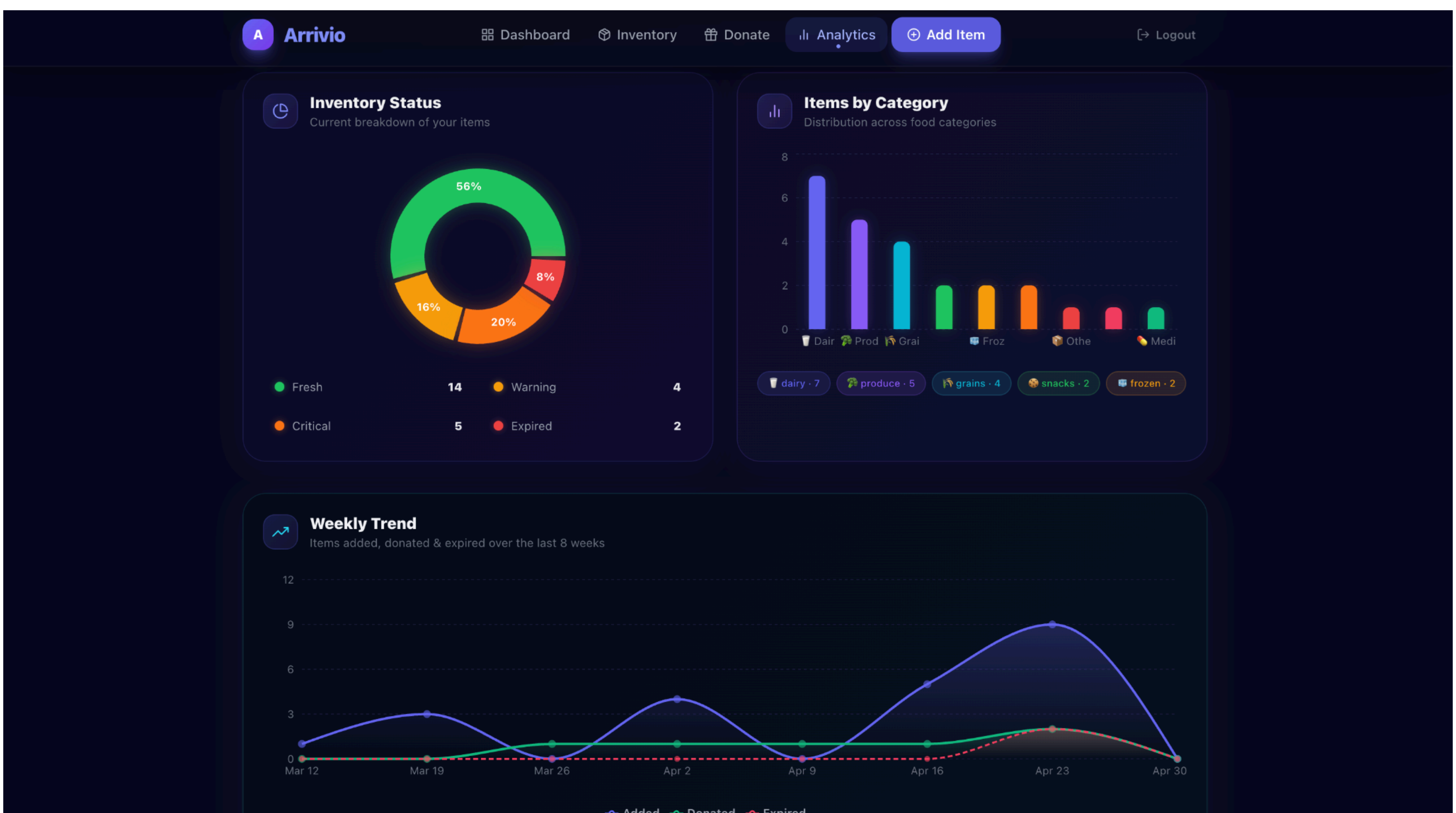
4.1 Tools and Technologies Used

Layer	Technology	Version	Purpose
Frontend Framework	React.js	18.3.1	SPA UI rendering
State Management	Redux Toolkit	2.2.7	Global state
Routing	React Router DOM	6.26.1	Client-side navigation
Styling	Tailwind CSS	3.4.10	Utility-first dark theme
Build Tool	Vite	5.4.11	Fast dev server & bundler
HTTP Client	Axios	1.7.7	REST API calls

Layer	Technology	Version	Purpose
Charts	**Recharts**	**2.x**	**Analytics visualisations [NEW]**
Backend Runtime	Node.js	20.x LTS	Server-side JavaScript
Web Framework	Express.js	4.21.2	REST API server
Database	MongoDB	7.x	NoSQL document storage
ODM	Mongoose	8.17.0	Schema + query abstraction
Authentication	JWT	9.0.2	Stateless auth via cookies
Password Hashing	bcrypt	6.0.0	Secure password storage
Geolocation	Browser Geolocation API	Native	Near Me NGO detection




feeding india
A ZOMATO GIVEBACK
Our work
About us
Partner with us
Donate now
Nourished children are the foundation of a thriving India.
Scroll down

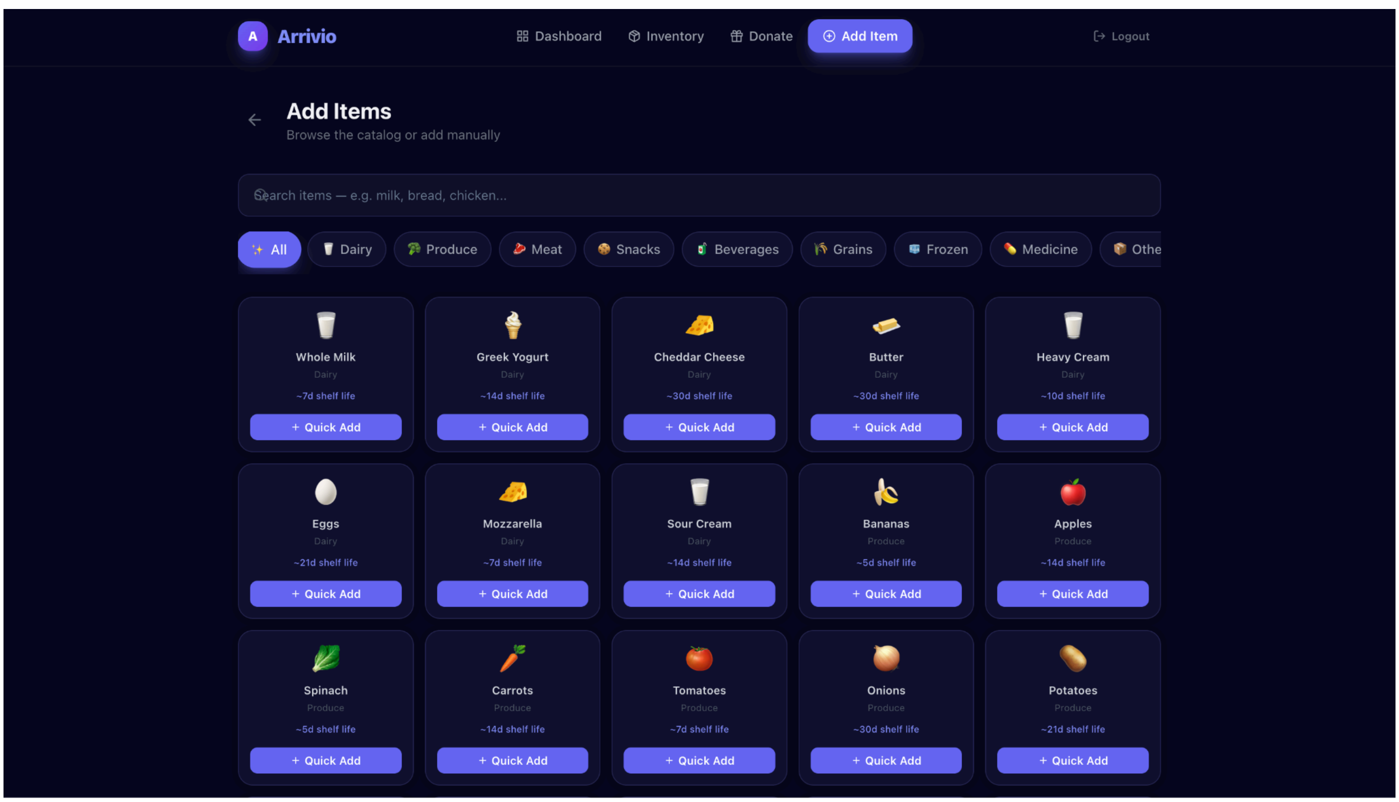

A
Arrivio
Dashboard
Inventory
Donate
Analytics
Add Item
Logout
Inventory Status
Current breakdown of your items
56%
8%
16%
20%
Fresh 14
Warning 4
Critical 5
Expired 2
Items by Category
Distribution across food categories
8
6
4
2
0
Dair Prod Grai Froz Othe Medi
dairy · 7
produce · 5
grains · 4
snacks · 2
frozen · 2
Weekly Trend
Items added, donated & expired over the last 8 weeks
12
9
6
3
0
Mar 12 Mar 19 Mar 26 Apr 2 Apr 9 Apr 16 Apr 23 Apr 30

4.2 Dataset Description

a) Predefined Item Catalog (Static Dataset — 55+ Items)

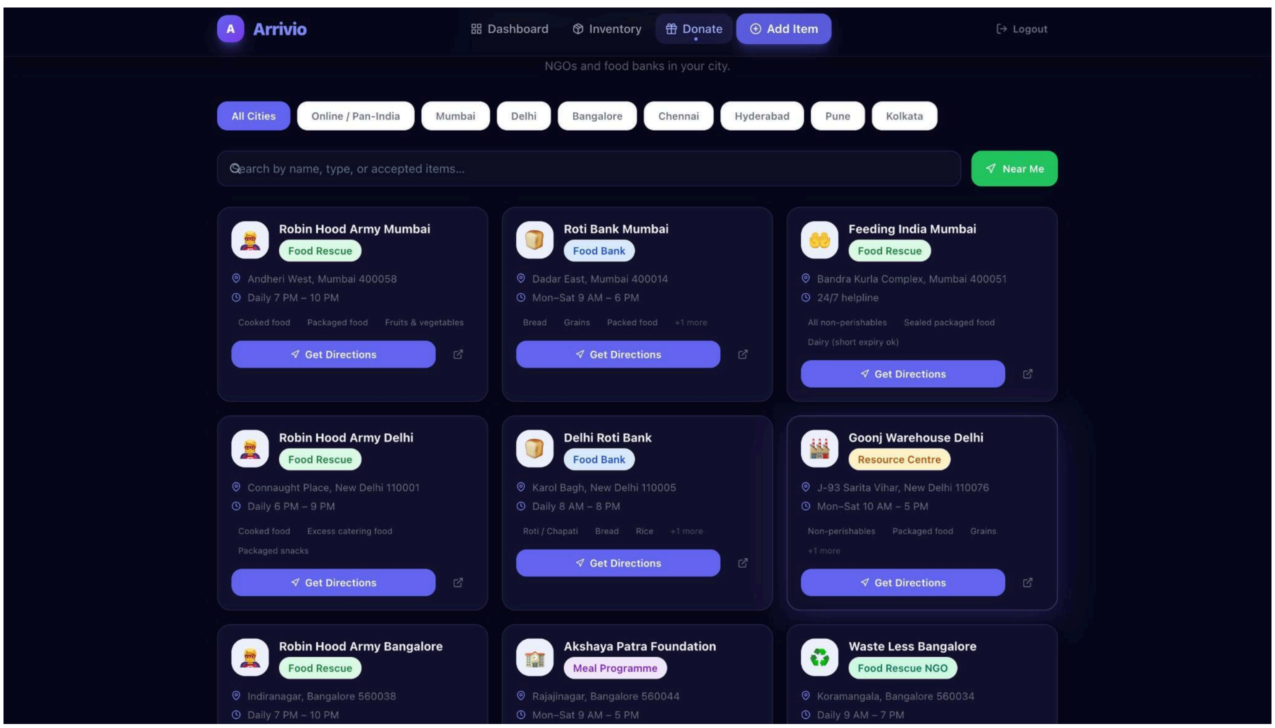
A curated catalog of 55+ common household items across 9 categories, each with pre-filled default values:

Category	Sample Items	Avg Shelf Life	Default Unit
Dairy	Milk, Yogurt, Cheese, Butter, Eggs	7–30 days	L / g / pcs
Produce	Apples, Bananas, Carrots, Spinach	3–14 days	pcs / g
Grains	Bread, Rice, Pasta, Oats	5–365 days	pcs / g
Meat	Chicken, Fish, Beef, Paneer	2–7 days	kg / g
Beverages	Juice, Water, Soda, Tea	7–365 days	L / mL
Snacks	Biscuits, Granola Bars, Chips	30–180 days	g / pcs
Frozen	Ice Cream, Frozen Peas, Pizza	30–365 days	g / pcs
Medicine	Vitamins, Paracetamol, Antacids	180–730 days	pcs
Other	Sauces, Condiments, Oils	30–730 days	mL / g



b) NGO Directory (Static Dataset — 30+ NGOs, 8 Indian Cities)

City	NGOs Listed	Types Covered
Mumbai	5	Food Rescue, Food Bank, Meal Programme
Delhi	5	Food Rescue NGO, National Network
Bangalore	4	Resource Centre, Charitable Organisation
Chennai	4	Food Bank, Meal Programme
Hyderabad	3	Food Rescue, Charitable Organisation
Pune	3	Food Bank, NGO
Kolkata	3	Charitable Organisation, Food Bank
Pan-India	3	National Programme, National Network



5. Conclusion

6.1 Summary of Contributions

Arrivio now delivers a complete, free, unified platform combining:

1. **Four-tier colour-coded expiry intelligence** — Fresh / Warning / Critical / Expired
2. **Real-time inventory health ring** — household freshness as a live percentage
3. **Peer-to-peer community donation board** — list and claim near-expiry items

4. **Geolocation-integrated NGO finder** — 30+ verified food banks, 8 Indian cities
5. **Animated dark-theme UI** — 4.9/5 visual appeal, low churn engagement design
6. **Carbon Footprint Tracker** *(new)* — per-user CO_2 savings quantified from real donation data
7. **Interactive Analytics Dashboard** *(new)* — four chart types backed by a dedicated API endpoint

Arrivio is, to the best of our knowledge, the first platform to combine expiry-based inventory tracking, peer-to-peer community donation, verified NGO integration, **and personal carbon footprint analytics** in a single free web application.

6.2 Social Value and Impact

Scale	**Impact Projection**
Household (1,000 families)	20% waste reduction = ~20,000 kg food saved annually
Community	67% claim rate = food redistribution addressing food insecurity
Environmental	Each kg prevented = ~2.5 kg CO_2 eliminated [EPA, 2023]
NGO	Digital connectivity increases intake without additional outreach
Individual (new)	**CO_2 tracker creates personal accountability and behaviour feedback loop**

6.3 Future Work (Remaining)

The following enhancements are planned for future versions:

#	**Feature**	**Priority**
1	**Barcode Scanning** — camera-based automatic item entry	High
2	**Mobile Application (React Native)** — iOS & Android with push notifications	High
3	**AI-powered Shopping Planner** — suggest grocery lists from purchase history	Medium
4	**Smart Expiry Prediction** — ML model trained on item type, storage, climate	Medium

#	Feature	Priority
5	**Multi-language Support** — Hindi, Tamil, Telugu for wider accessibility	Medium
6	**SMS Alert System** — expiry alerts via SMS for users without internet	Medium
7	**Expanded NGO Database** — grow from 8 to 100+ Indian cities	Low
8	**Gamification Layer** — badges, leaderboards, community challenges	Low
9	**Government API Integration** — FSSAI databases for real-time shelf-life standards	Low

Note: Carbon Footprint Tracker and Waste Analytics Dashboard — originally listed as Future Work in Evaluation 2 — have been **fully implemented** in Evaluation 3.

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3. Schanes, K., Dobernig, K., & Gözet, B. (2018). Food waste matters. *Journal of Cleaner Production*, 182, 978–991.
4. Aschemann-Witzel, J. et al. (2015). Consumer-related food waste. *Sustainability*, 7(6), 6457–6477.
5. Stancu, V., Haugaard, P., & Lähteenmäki, L. (2016). Determinants of consumer food waste. *Appetite*, 96, 7–17.
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12. van Herpen, E. et al. (2019). Comparing wasted food with food tracking app use. *Resources, Conservation and Recycling*, 148, 73–84.
